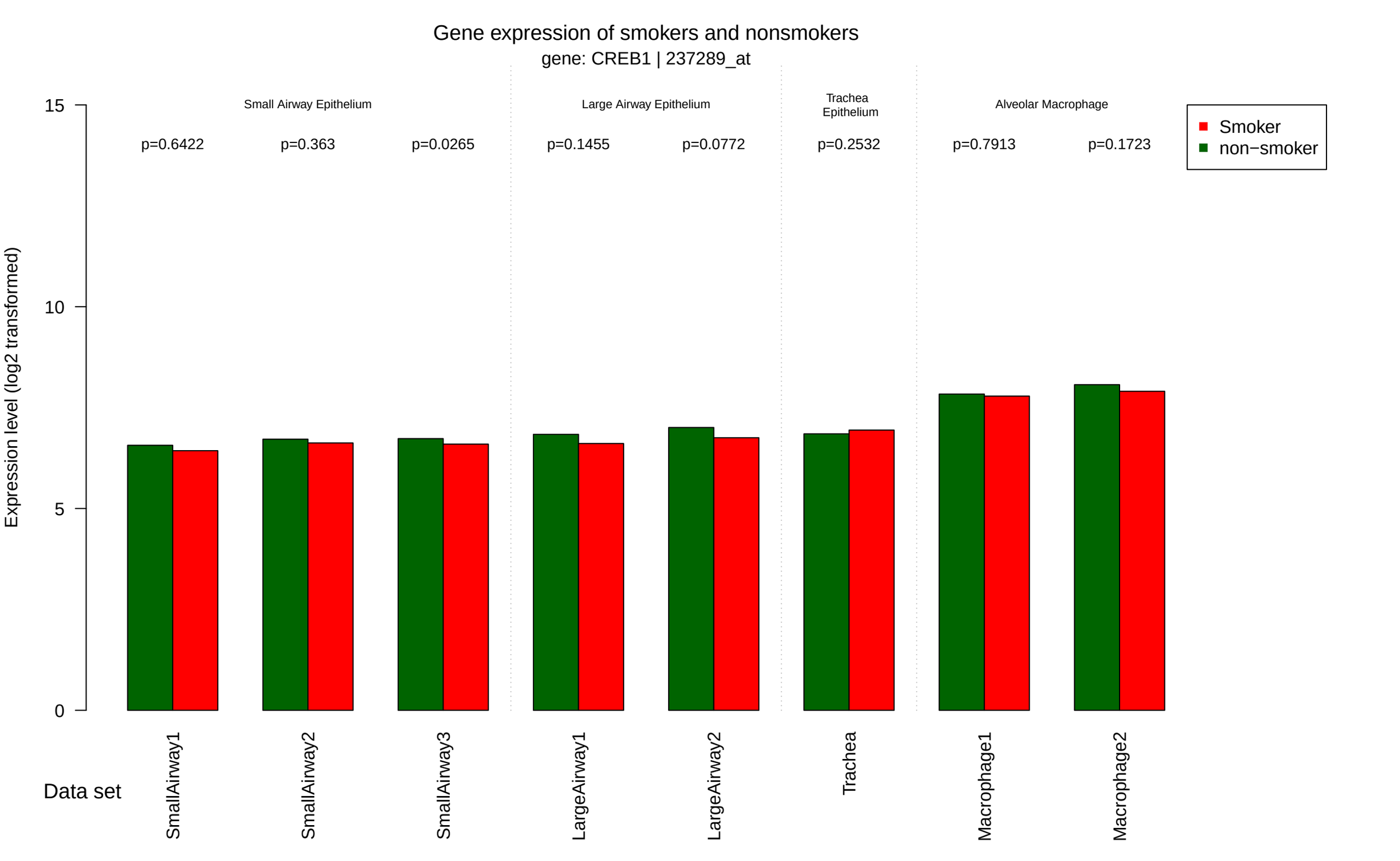
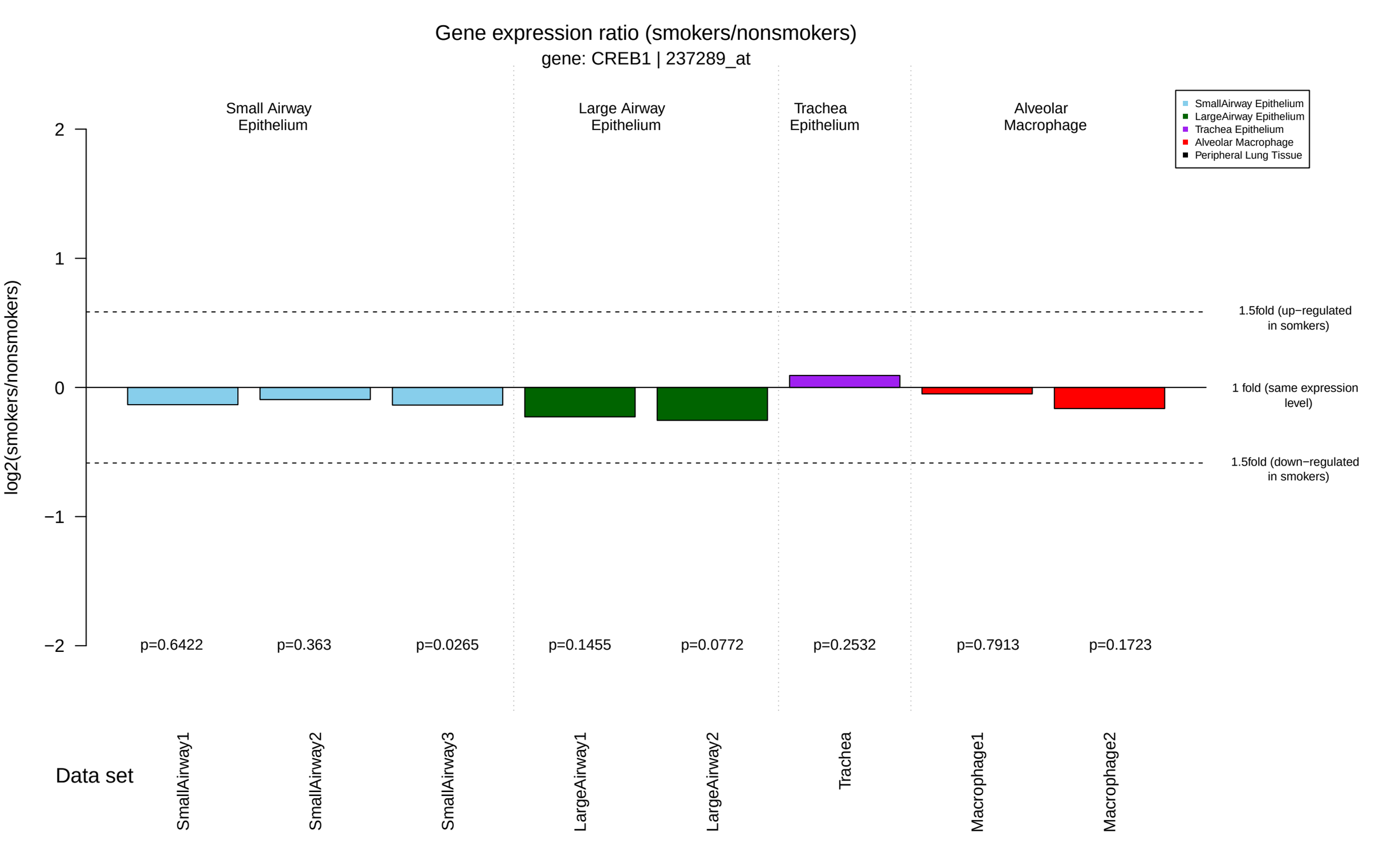
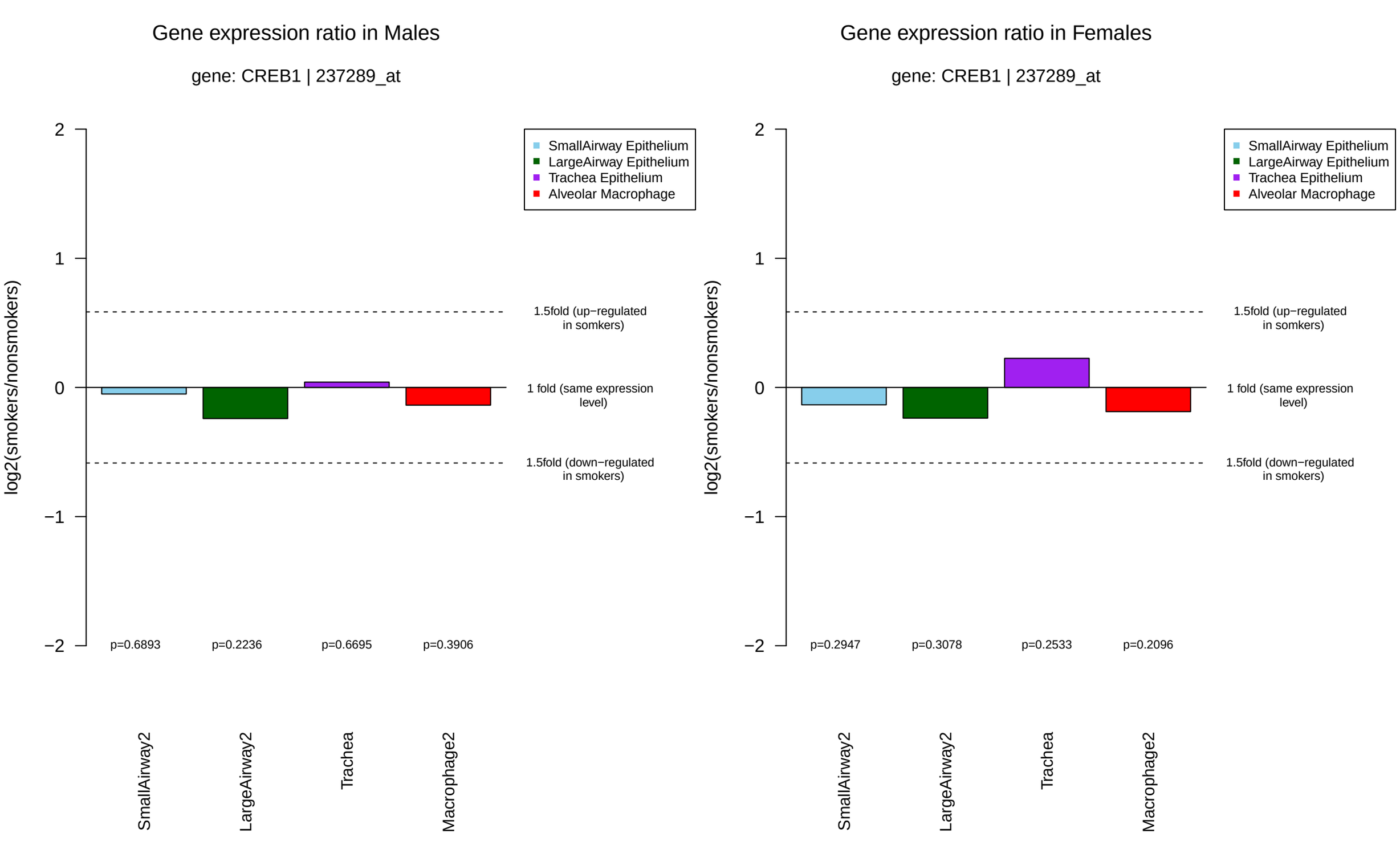
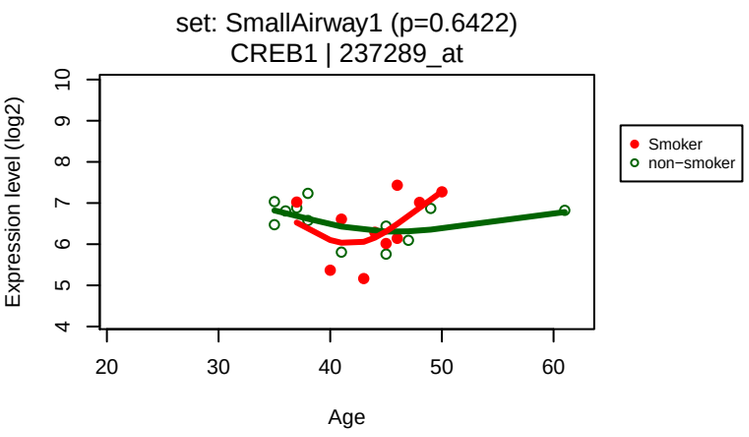
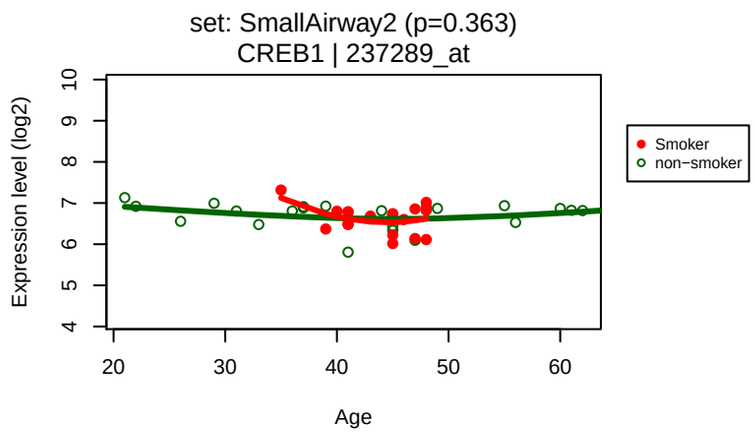
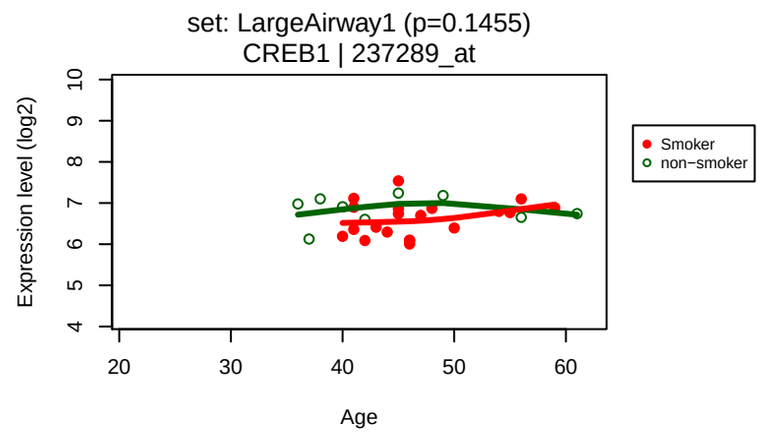

Gene expression of smokers and nonsmokers
gene: CREB1 | 237289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6422
p=0.363
p=0.0265
p=0.1455
p=0.0772
p=0.2532
p=0.7913
p=0.1723
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

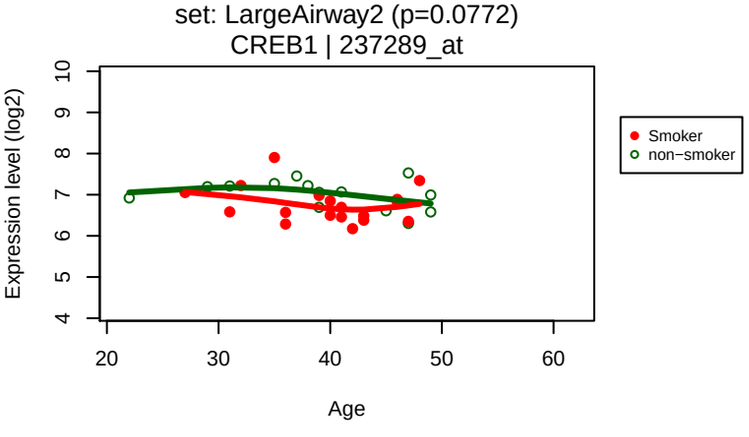

Gene expression ratio (smokers/nonsmokers)
gene: CREB1 | 237289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6422
p=0.363
p=0.0265
p=0.1455
p=0.0772
p=0.2532
p=0.7913
p=0.1723
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

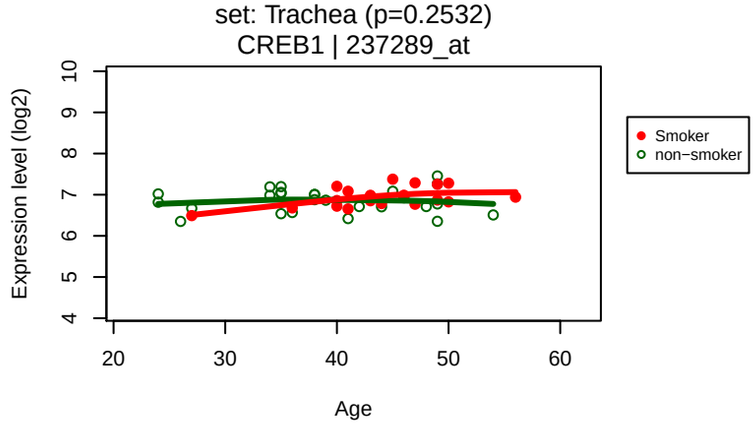

Gene expression ratio in Males
gene: CREB1 | 237289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6893
p=0.2236
p=0.6695
p=0.3906
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB1 | 237289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2947
p=0.3078
p=0.2533
p=0.2096
SmallAirway2
LargeAirway2
Trachea
Macrophage2

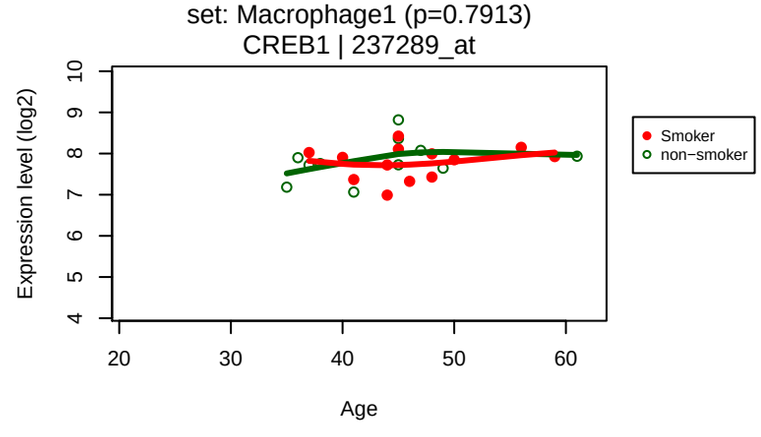

set: SmallAirway1 (p=0.6422)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker

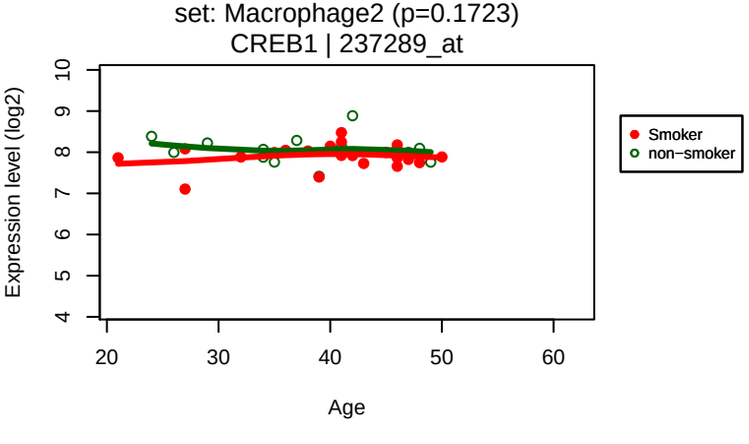

set: SmallAirway2 (p=0.363)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker

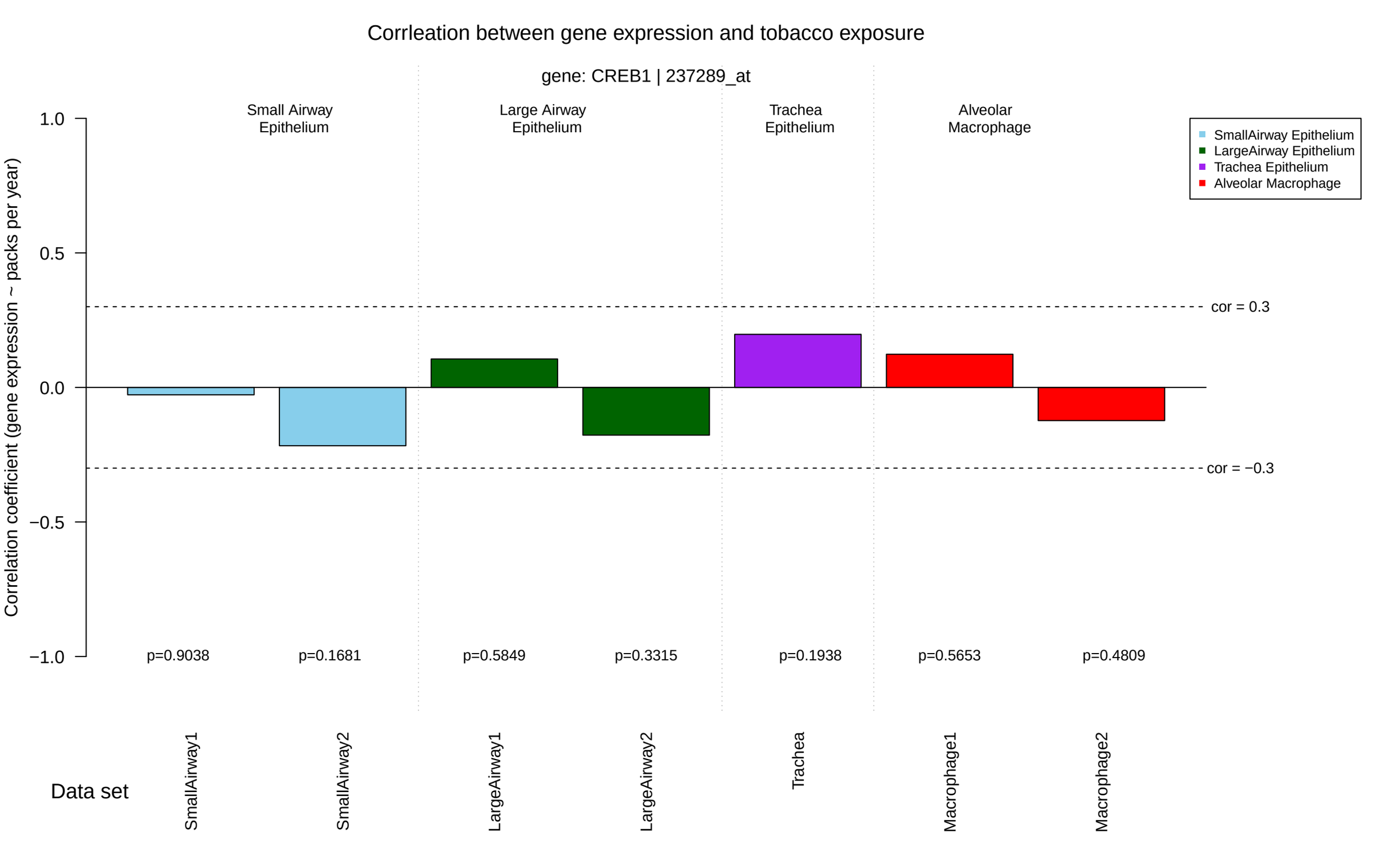

set: LargeAirway1 (p=0.1455)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0772)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2532)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7913)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1723)
CREB1 | 237289_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB1 | 237289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9038
p=0.1681
p=0.5849
p=0.3315
p=0.1938
p=0.5653
p=0.4809
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2